INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA
SALT LAKE
(Affiliated to UEM Kolkata and Approved by AICTE)

Advanced Programming Lab
[PCCCS495]

Student Attendance Management System

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Semester: -IV

Class Roll: - 64

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2. Problem Statement

Managing library records manually is inefficient and prone to errors. Traditional systems using registers or spreadsheets make it difficult to track book availability, issue records, and returns.

This project aims to develop a console-based Library Management System using Java, which allows users to:

- Add books
- Display available books
- Issue books
- Return books

The goal is to create a structured and efficient system using Object-Oriented Programming (OOP) principles.

3. System Design

3.1 Class Diagram (Textual Representation)

- Book
- Library
- Main

3.2 Description of Major Classes

i. Book Class

- Represents a book entity
- Attributes: id, title, author, isIssued
- Methods: issueBook(), returnBook(), display()

ii. Library Class

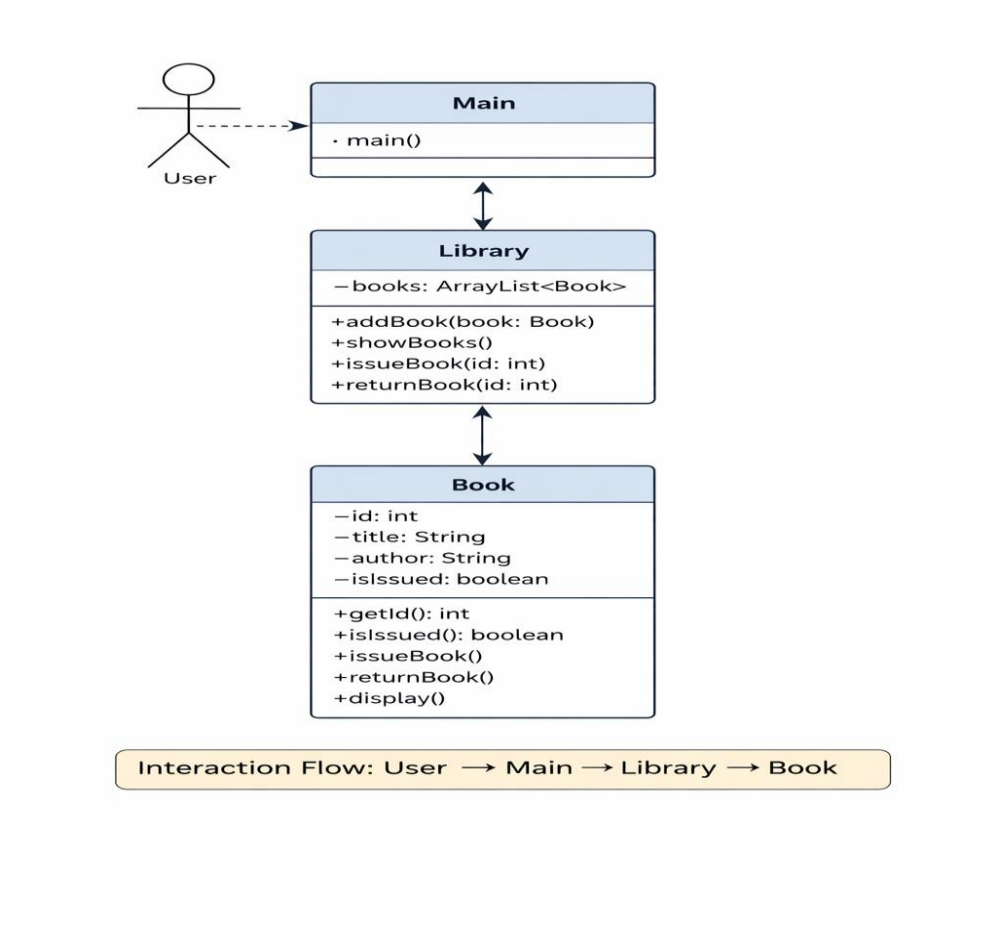
- Manages collection of books using ArrayList
- Handles adding, issuing, returning, and displaying books

iii. Main Class

- Entry point of the program
- Provides menu-driven interface for user interaction

3.4 Interaction Flow

1. User starts the program
2. Menu is displayed
3. User selects an option:
    - Add book → stored in list
    - Show books → display all books
    - Issue book → update status
    - Return book → update status
4. Program continues until user exits



4. OOP Concepts Demonstrated

4.1 Encapsulation

- Used in Book class with private variables and public methods
- Ensures data protection

4.2 Abstraction

- Hides internal implementation of book operations
- User interacts only through methods

4.3 Polymorphism

- display() method shows different outputs based on book status

4.4 Collections

- ArrayList is used to store and manage books dynamically

5. Implementation Highlights

5.1 Adding Books

books.add(book);
Ensures dynamic storage of book records.

5.2 Issue Book Logic

if (b.getId() == id && !b.isIssued())
Prevents issuing already issued books.

5.3 Return Book Logic

if (b.getId() == id && b.isIssued())
Ensures only issued books can be returned.

5.4 Menu-driven Interaction

- Uses switch-case for user choices
- Provides simple and user-friendly interface

6. Testing & Error Handling

6.1 Test Cases

Test Case	Input	Output
Add Book	Valid data	Book added
Issue Book	Valid ID	Book issued
Return Book	Valid ID	Book returned
Invalid ID	Wrong ID	Error message

6.2 Edge Cases Handled

- No books available
- Invalid book ID
- Re-issuing already issued book
- Returning non-issued book

6.3 Failure Handling

- Invalid menu choice → handled with default case
- Logical errors → handled using conditions

7. Git Workflow:

Commits

History for term-ii-project-submission-sulagnaghoshofficial-ui / README.md on main

All users | All time

Commits on Mar 29, 2026

- architecture added — sulagnaghoshofficial-ui committed 2 hours ago — 41b55dc
- how to run section is added — sulagnaghoshofficial-ui committed 8 hours ago — 11c5d4e
- added — sulagnaghoshofficial-ui committed 9 hours ago — a112275
- readme file is edited — sulagnaghoshofficial-ui committed 9 hours ago — 8af021a

Commits on Mar 28, 2026

- Update README.md — sulagnaghoshofficial-ui authored yesterday — Verified — cc51ca3

Commits on Mar 25, 2026

- core feature is added — sulagnaghoshofficial-ui committed 4 days ago — b2bd2ed
- target user description added — sulagnaghoshofficial-ui committed 4 days ago — fed06aa

History for term-ii-project-submission-sulagnaghoshofficial-ui / src on main

All users | All time

Commits on Mar 29, 2026

- main class added — sulagnaghoshofficial-ui committed 9 hours ago — 67eddcc
- added library class — sulagnaghoshofficial-ui committed 9 hours ago — c1d9b25
- Created Library class to manage collection of books — sulagnaghoshofficial-ui committed 9 hours ago — 222d05f

Commits on Mar 28, 2026

- added book class — sulagnaghoshofficial-ui committed yesterday — 68909b9

Commits on Mar 25, 2026

- target user section added — sulagnaghoshofficial-ui committed 4 days ago — f7247ec
- oops project — sulagnaghoshofficial-ui committed 4 days ago — 2359988

Commits on Mar 2, 2026

- Initial commit — github-classroom[bot] authored 27 days ago — Verified — 59939a0

End of commit history for this file

- readme file is edited — sulagnaghoshofficial-ui committed 9 hours ago — 8af021a

Commits on Mar 28, 2026

- Update README.md — sulagnaghoshofficial-ui authored yesterday — Verified — cc51ca3

Commits on Mar 25, 2026

- core feature is added — sulagnaghoshofficial-ui committed 4 days ago — b2bd2ed
- target user description added — sulagnaghoshofficial-ui committed 4 days ago — fed06aa

Commits on Mar 23, 2026

- Update README.md — sulagnaghoshofficial-ui authored last week — Verified — be60f66

Commits on Mar 19, 2026

- add deadline — github-classroom[bot] authored last week — Verified — 174af25

Commits on Mar 2, 2026

- Initial commit — github-classroom[bot] authored 27 days ago — Verified — 59939a0

End of commit history for this file

8. Conclusion & Future Scope

The Library Management System successfully demonstrates the use of OOP concepts in building a structured and efficient application. It simplifies book management through a user-friendly console interface.

Future Enhancements:

- Graphical User Interface (GUI)
- Database integration (MySQL)
- User authentication system
- Fine calculation system
- Search and filter functionality